EmPath Team Work Plan

Scope: Plano ZIP areas 75025, 75024, 75023, and 75074.

- Nalini — 20 hours: implementation, routing, map, and presentation.
- Pramukh — 15 hours: GIS review, model evaluation, and troubleshooting.
- Both totals include the same two-hour team meeting.

1 Before the Hackathon

Nalini — 9 hours

- 1 hour: Confirm Python setup; complete sources and coverage checklist.
- 2 hours: Clean boundaries, walking data, residential origins, and destinations.
- 2.5 hours: Build walking graph; connect origins and destinations.
- 2 hours: Add canopy and environmental features.
- 1.5 hours: Select and test an existing GeoAI model or workflow.

Pramukh — 4 flexible hours outside the hackathon

- 1 hour: Review roadmap, data coverage, and model approach.
- 3 hours: Fix a priority network/data gap, reproduce the model workflow, or verify entrances and crossings.
- Return reviewed files, documented fixes, and unresolved issues. Schedule when available; Nalini continues independently if needed.

2 Shared Team Meeting

Two hours each, already included in both budgets.

- 30 minutes: Confirm scope and minimum demo.
- 30 minutes: Review walking graph and data gaps.
- 30 minutes: Agree on GeoAI approach and evaluation.
- 30 minutes: Assign tasks and handoffs. Save one shared task list.

3 Hackathon

Nine hours each. Nalini reserves hours 6–9 for packaging and presentation.

Hours 0 to 4

- Nalini: Finish GeoAI adapter; calculate routes and residential rankings.
- Pramukh: Audit GIS layers, connections, entrances, and at least one route per ZIP.

Hours 4 to 7

- Nalini: Finish the map and priority fixes by hour 6; then begin packaging and slides.
- Pramukh: Check model outputs; compare distance-only and friction routes; test two penalty settings.

Hours 7 to 9

- Nalini: Finalize demo, README, backup, and five-minute presentation.
- Pramukh: Resolve final issues, verify results, and rehearse.

4 Required Deliverables

- Nalini: Sources checklist; clean data; walking graph and features; runnable GeoAI component; routes and rankings; interactive map; README; slides and backup.
- Pramukh: Network audit with flagged locations and fixes; model/route evaluation; limitations and completed checks.

5 Optional Work for Pramukh

Only after the planned 15 hours, if more time is available.

- 1 hour: Verify additional routes and destinations.
- 1–2 hours: Test how penalty settings affect residential rankings.
- 2–3 hours: Compare a second existing model if compatible data and labels are available.
- Prioritize fixes and validation before new features.

6 Working Rules

- Exchange GraphML, GeoPackage, and CSV files with stable IDs and documented units.
- Pramukh may review in Python or ArcGIS Pro; Nalini maintains the Python pipeline.
- Track issues in one list: Problem → Owner → Fix → Status.
- If behind, reduce sampled homes or optional layers; retain all four ZIP areas.

Time Check

- Nalini: 9 hours preparation + 2 hours meeting + 9 hours hackathon = 20 hours.
- Pramukh: 4 flexible hours + 2 hours meeting + 9 hours hackathon = 15 hours.